# Experiment No. 2

Aim:- Create blockchain using python

## What is Blockchain?

Blockchain is a modern digital technology used to store information in a **secure, transparent, and tamper-proof way**. It works like a **shared online record book (ledger)** that keeps track of transactions.

The term **blockchain** comes from how the data is stored:

- Information is saved inside **blocks**
- Each block is connected to the **previous block**
- Together they form a **chain of blocks**

Every block contains:

- Transaction details
- Time of creation (timestamp)
- A special security code called a **cryptographic hash**

This hash links one block to another. If someone tries to change the data, the hash changes and the chain breaks, making tampering easy to detect.

## Key Features of Blockchain

- Distributed across many computers (nodes)
- No single authority controls it (decentralized)
- Transactions are verified by the majority of nodes
- Once data is added, it cannot be changed (immutable)
- Highly secure and transparent

In simple words, blockchain is a **decentralized database that safely records transactions across multiple systems**.

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# What is a Block?

A **block** is a basic unit of storage in a blockchain.
Think of it like **one page in a ledger book** that stores transaction records.

Many transactions happen daily, and blocks help organize and store these transactions securely. Once a block is full, it gets added to the blockchain and linked with other blocks.

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# Components of a Block

## Block Header

The block header contains important information that uniquely identifies the block. It helps maintain the order and connection between blocks.

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## Previous Block Hash

This stores the hash of the previous block.
It links the current block to the earlier one, forming the **chain structure** of the blockchain.

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## Timestamp

Records the exact **date and time** when the block was created.
It proves when the transactions were added.

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## Nonce

Nonce means **“number used only once.”**

- Used during mining (Proof of Work)
- Miners keep changing the nonce to find a valid hash
- Helps secure the block and validate it

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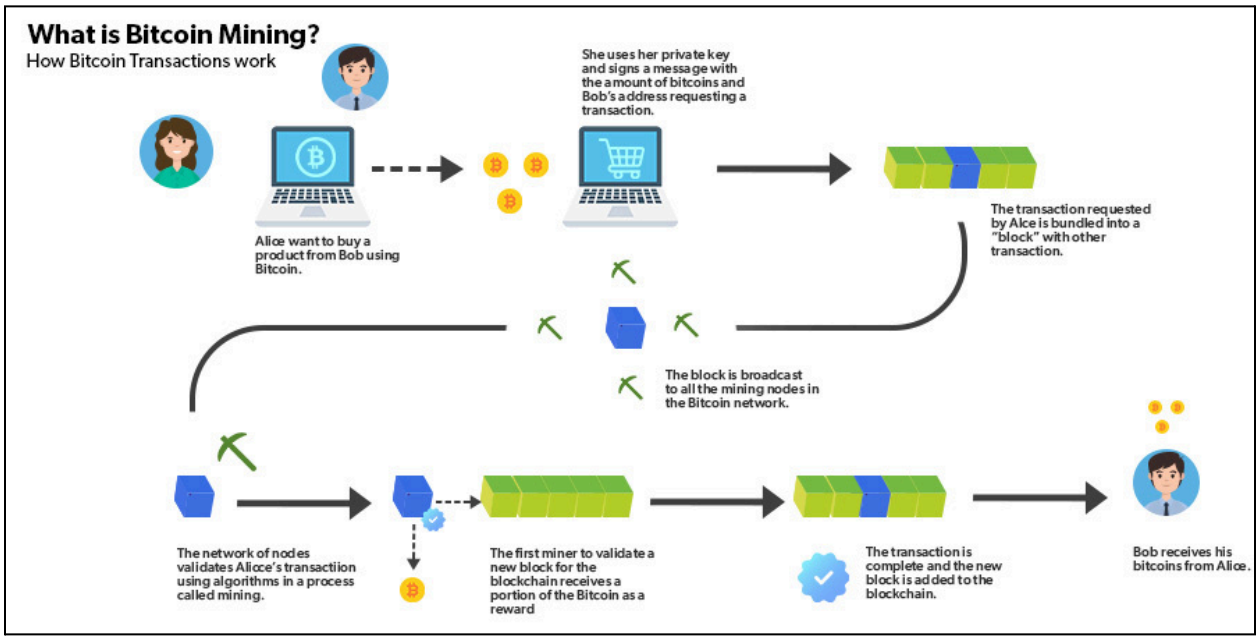
## Merkle Root

The Merkle Root is created using a **Merkle Tree**.

- Combines all transactions into one hash
- Acts like a digital fingerprint
- Allows quick and secure verification of transactions


```
Block {
  Index: 2
  Timestamp: 2026-01-22 10:05:00
  Data: {"Sender":"Alice", "Receiver":"Bob", "Amount":50}
  Nonce: 102345
  Previous Hash: "0000a1b2c3d4e5..."
  Hash: "0000f3b7d6c1e8..."
}
```

Process of mining




Mining is the process of adding a new block to the blockchain.

It ensures security and integrity using Proof-of-Work (PoW).

Step 1: Collect Transactions

- All pending transactions are collected into a new block.
- Example: Alice sends 50 coins to Bob.

Step 2: Create Block Header

Block header includes:

- Previous hash
- Transactions
- Timestamp
- Nonce (initially 0)

Step 3: Proof of Work (PoW)

- The miner repeatedly changes the nonce to find a hash that satisfies the difficulty requirement.
- Difficulty = number of leading zeroes required in hash (e.g., 4 leading zeroes).

SHA256(block_data + nonce) → hash starts with "0000"

- This requires computational work, hence the name Proof-of-Work.

Step 4: Validate and Broadcast

- Once a valid hash is found, the block is broadcast to all nodes in the network.
- Other nodes verify:

    1. Hash correctness
    2. Nonce validity
    3. Previous hash link

Step 5: Add to Blockchain

- If all nodes validate, the block is added to the chain.
- The miner receives a reward (e.g., cryptocurrency).

How to check validity of blocks in blockchain

Step 1: Validate Previous Hash

- Each block's previous_hash must match the hash of the previous block.

```
if Block[n].previous_hash != Hash(Block[n-1]):
    Invalid
```

Step 2: Recalculate Current Hash

- Compute hash of current block's data and nonce.
- It must match the stored hash in the block.

```
if SHA256(Block[n].data + Block[n].nonce) != Block[n].hash:
    Invalid
```

Step 3: Check Proof-of-Work

- Verify that the block hash satisfies the difficulty (e.g., 4 leading zeroes).

```
if not Block[n].hash.startswith("0000"):
    Invalid
```

Step 4: Validate Entire Chain

- Repeat steps 1–3 for all blocks from Genesis to the latest block.
- If any block fails → blockchain is invalid.

Step 5: Validate Genesis Block

- The first block (Genesis block) is hardcoded.
- Previous hash = "0"
- Must not be altered.

```
PS C:\NRK> python -m pip install flask
Collecting flask
  Downloading flask-3.1.2-py3-none-any.whl.metadata (3.2 kB)
Collecting blinker>=1.9.0 (from flask)
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click>=8.1.3 (from flask)
  Downloading click-8.3.1-py3-none-any.whl.metadata (2.6 kB)
Collecting itsdangerous>=2.2.0 (from flask)
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting jinja2>=3.1.2 (from flask)
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting markupsafe>=2.1.1 (from flask)
  Downloading markupsafe-3.0.3-cp314-cp314-win_amd64.whl.metadata (2.8 kB)
Collecting werkzeug>=3.1.0 (from flask)
  Downloading werkzeug-3.1.5-py3-none-any.whl.metadata (4.0 kB)
Collecting colorama (from click>=8.1.3->flask)
  Downloading colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Downloading flask-3.1.2-py3-none-any.whl (103 kB)
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Downloading click-8.3.1-py3-none-any.whl (108 kB)
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
Downloading markupsafe-3.0.3-cp314-cp314-win_amd64.whl (15 kB)
Downloading werkzeug-3.1.5-py3-none-any.whl (225 kB)
```

```
Installing collected packages: markupsafe, itsdangerous, colorama, blinker, werkzeug, jinja2, click, flask
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 7/8 [flask]  WARNING: The script flask.exe is installed in 'C:\Us
rs\admin\AppData\Local\Python\pythoncore-3.14-64\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-loc
tion.
Successfully installed blinker-1.9.0 click-8.3.1 colorama-0.4.6 flask-3.1.2 itsdangerous-2.2.0 jinja2-3.1.6 m
rkupsafe-3.0.3 werkzeug-3.1.5
```

**blockchain.py**

```
import datetime
import hashlib
import json


class Blockchain:
    def __init__(self):
        self.chain = []
        self.difficulty = 4  # 4 leading zeroes
        self.create_genesis_block()

    def create_genesis_block(self):
        genesis_block = {
```

```
        'index': 1,
        'timestamp': str(datetime.datetime.now()),
        'data': 'Genesis Block',
        'nonce': 0,
        'previous_hash': '0'
    }
    genesis_block['hash'] = self.calculate_hash(genesis_block)
    self.chain.append(genesis_block)

def calculate_hash(self, block):
    block_copy = block.copy()
    block_copy.pop('hash', None)
    encoded = json.dumps(block_copy, sort_keys=True).encode()
    return hashlib.sha256(encoded).hexdigest()

def mine_block(self, data):
    previous_block = self.chain[-1]
    nonce = 0

    while True:
        block = {
            'index': len(self.chain) + 1,
            'timestamp': str(datetime.datetime.now()),
            'data': data,
            'nonce': nonce,
            'previous_hash': previous_block['hash']
        }

        block_hash = self.calculate_hash(block)

        if block_hash.startswith('0' * self.difficulty):
            block['hash'] = block_hash
            self.chain.append(block)
            return block

        nonce += 1

def is_chain_valid(self):
    for i in range(1, len(self.chain)):
        current = self.chain[i]
        previous = self.chain[i - 1]

        if current['previous_hash'] != previous['hash']:
            return False
```

```
        recalculated_hash = self.calculate_hash(current)
        if recalculated_hash != current['hash']:
            return False

        if not current['hash'].startswith('0' * self.difficulty):
            return False

    return True
```

**app.py**

```
from flask import Flask, jsonify
from blockchain import Blockchain

app = Flask(__name__)
blockchain = Blockchain()


@app.route('/mine_block', methods=['GET'])
def mine_block():
    block = blockchain.mine_block("Some transaction data")

    return jsonify({
        'message': '⛏️ Block mined successfully!',
        'block': block
    }), 200


@app.route('/get_chain', methods=['GET'])
def get_chain():
    return jsonify({
        'chain': blockchain.chain,
        'length': len(blockchain.chain)
    }), 200


@app.route('/is_valid', methods=['GET'])
def is_valid():
    return jsonify({
        'is_valid': blockchain.is_chain_valid()
    }), 200
```

```
if __name__ == '__main__':
    print(" Mining blockchain with 4 leading zeroes...")
    app.run(debug=True, use_reloader=False)
```

127.0.0.1:5000/mine_block

Pretty print


```
{
  "block": {
    "data": "Some transaction data",
    "hash": "00002bbdd893c67f29a8d2ec5e4fe64e4c8cbc836084bb3ca6944b2a60f29e69",
    "index": 4,
    "nonce": 58439,
    "previous_hash": "00007957030d4e8b2432e5037939f03d9b16d8612364ce8c6cefd2b3e2a0d5f6",
    "timestamp": "2026-01-22 12:06:09.082474"
  },
  "message": "\u26cf\ufe0f Block mined successfully!"
}
```


127.0.0.1:5000/is_valid

Pretty print


```
{
  "is_valid": true
}
```

127.0.0.1:5000/get_chain

Pretty print


```
{
  "chain": [
    {
      "data": "Genesis Block",
      "hash": "d1c793d584710b37ed90d397f2f7a59cd9f741b1f424553cae1948dfeed661c4",
      "index": 1,
      "nonce": 0,
      "previous_hash": "0",
      "timestamp": "2026-01-22 12:05:29.544918"
    },
    {
      "data": "Some transaction data",
      "hash": "000030b5874b1e81cad31a1eb57cbcf69db993bbb0af36834c9742c4bea27ab5",
      "index": 2,
      "nonce": 215962,
      "previous_hash": "d1c793d584710b37ed90d397f2f7a59cd9f741b1f424553cae1948dfeed661c4",
      "timestamp": "2026-01-22 12:05:49.656127"
    },
    {
      "data": "Some transaction data",
      "hash": "00007957030d4e8b2432e5037939f03d9b16d8612364ce8c6cefd2b3e2a0d5f6",
      "index": 3,
      "nonce": 30433,
      "previous_hash": "000030b5874b1e81cad31a1eb57cbcf69db993bbb0af36834c9742c4bea27ab5",
      "timestamp": "2026-01-22 12:06:04.610851"
    },
    {
      "data": "Some transaction data",
      "hash": "00002bbdd893c67f29a8d2ec5e4fe64e4c8cbc836084bb3ca6944b2a60f29e69",
      "index": 4,
      "nonce": 58439,
      "previous_hash": "00007957030d4e8b2432e5037939f03d9b16d8612364ce8c6cefd2b3e2a0d5f6",
      "timestamp": "2026-01-22 12:06:09.082474"
    }
  ],
  "length": 4
}
```